

Recommendation Scores

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can compute a weighted score.

I can apply hard constraints before ranking.

I can explain how weight choices change outcomes.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 23: Recommendation Scores

A weighted score expresses priorities but also embeds choices.



THINKER MISSION

Create two user profiles and calculate book rankings.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Which is closer under absolute-gap distance: a gap of 2 or a gap of 5?

This week's end goal

By the end, I can rank only eligible options with a transparent weighted score.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use recommendation scores to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

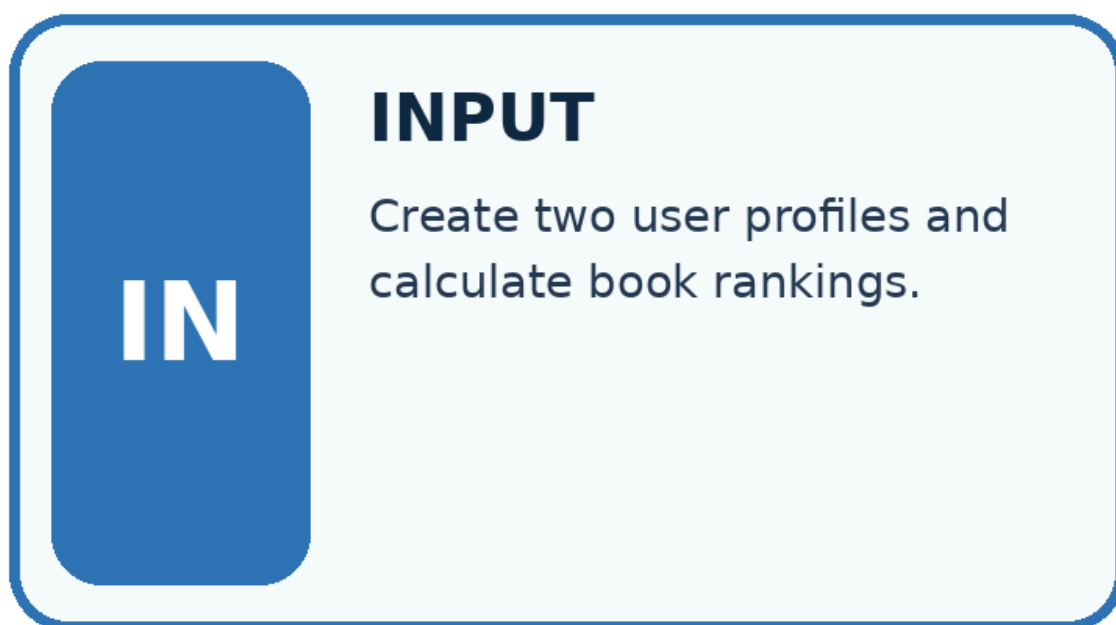
A fictional activity chooser uses enjoyment and learning ratings from 0 to 5. Its score is 2 times enjoyment plus learning. The activity must also take at most 20 minutes. Duration is a hard requirement; a high score cannot compensate for exceeding it.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



PROCESS

A weighted score expresses priorities but also embeds choices.



OUT

OUTPUT

Create two user profiles and calculate book rankings.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
options = [{"name": "A", "fun": 4, "learning": 2, "minutes": 15},
            {"name": "B", "fun": 3, "learning": 5, "minutes": 25}]

for item in options:
    if item["minutes"] <= 20:
        score = 2 * item["fun"] + item["learning"]
        print(item["name"], score)
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: A

Evidence: $2 * 4 + 2 = 10$

Meaning: Eligible at 15 minutes

Step or case: B

Evidence: $2 * 3 + 5 = 11$

Meaning: Excluded at 25 minutes

Read each row from left to right and connect the evidence to the stated rule.

Only A is printed, with score 10. If B took 20 minutes it would qualify and have the higher score 11. The inclusive boundary matters. Ratings and weights are invented for practice and require justification before use in a real decision.

Check must-have rules before scoring

Make two decisions in order. First check the duration: A takes 15 minutes and qualifies under the 20-minute limit; B takes 25 minutes and does not. Then calculate scores for eligible choices. A scores 2 times 4 plus 2, or 10. B would score 11, but that score cannot remove the time requirement. Keep the rejection reason beside B instead of calling it a worse-scoring activity.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary

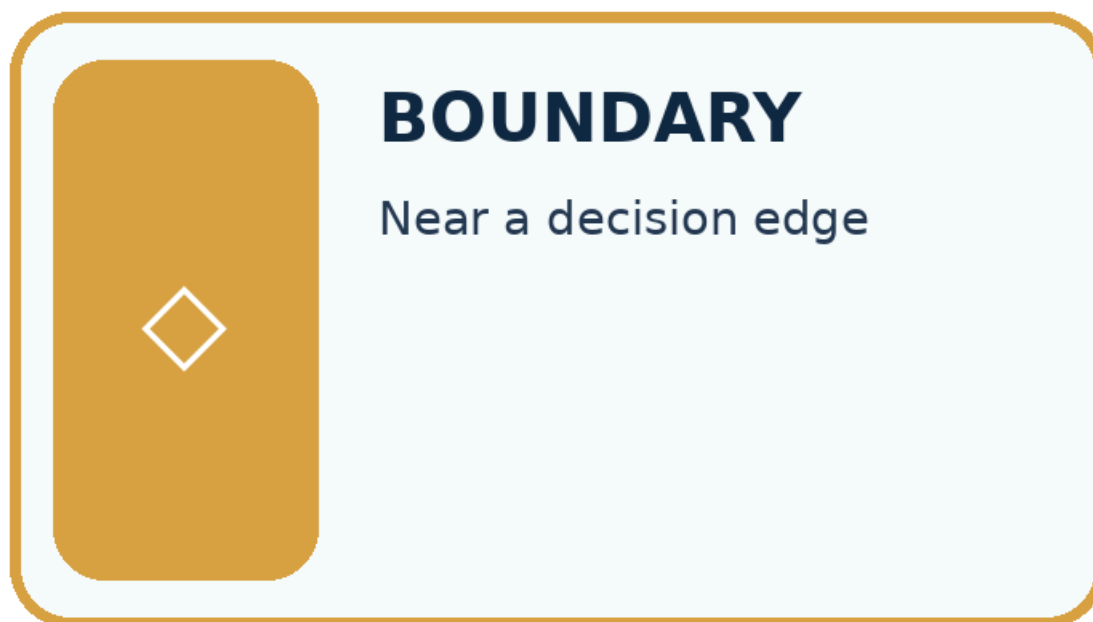
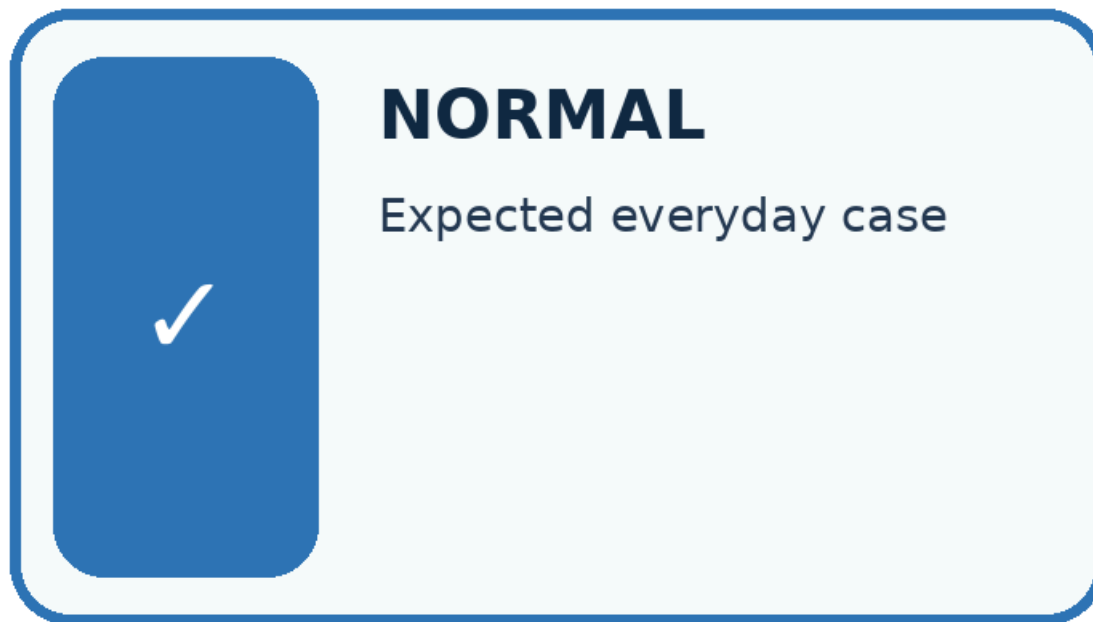
words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.

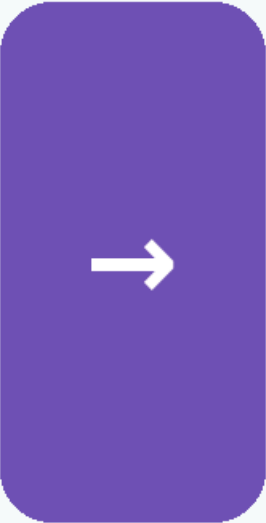


NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



TRANSFER

Fresh situation or user

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Change interest weight from 2 to 3, keeping time limit 14.

Prediction: A scores 11; C also scores 11; tie-by-ID chooses A.

Reason: B remains too long; changing preferences changes only ranking of A and C.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

SCORE

combined numeric evidence

3

RANK

order from score or rule

4

TRADEOFF

gain in one goal with cost in another

Word · My own example · Not this / boundary

Word: weight

My own example: A multiplier controlling how strongly a feature or value contributes to a calculation.

Not this / boundary: _____

Word: score

My own example: A numerical value assigned using a specified rule; it is not automatically a probability.

Not this / boundary: _____

Word: rank

My own example: Place candidates in an order using a declared comparison rule.

Not this / boundary: _____

Word: tradeoff

My own example: A situation where improving one desired property can make another harder to achieve.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Constraint

Meaning: A requirement an option must meet

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 23 • WORKED EVIDENCE

- 1 Score = $3 \times \text{genre match} + 2 \times \text{difficulty match} + 1 \times \text{length match}$.
- 2 Book A matches genre and length: $3 + 0 + 1 = 4$.
- 3 Book B matches difficulty and length: $0 + 2 + 1 = 3$.
- 4 A ranks first for these weights—not as a universal best book.

1. Score = $3 \times \text{genre match} + 2 \times \text{difficulty match} + 1 \times \text{length match}$.

2. Book A matches genre and length: $3 + 0 + 1 = 4$.

3. Book B matches difficulty and length: $0 + 2 + 1 = 3$.

4. A ranks first for these weights—not as a universal best book.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

If enjoyment weight becomes 1 and learning weight becomes 2, find both scores before applying the same duration limit.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Recommendation Scores

A fictional book-activity chooser has a 14-minute limit.

The facts we will use

Score = $2 \times \text{interest} + \text{practice}$. A = (interest 3, practice 2, time 14); B = (4, 3, 15); C = (2, 5, 12).

Require whole nonnegative feature values; time must be at most the limit. Highest score wins; tie by ID.

Time filtering and scores

	Minutes	Score	Status
A	14	8	eligible
B	15	11	excluded
C	12	9	eligible

Read a cell using both its row and column labels.

Recommendation Scores

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Eligibility first

A and C fit time; B takes 15 and is excluded.



2. Scores

A: $2 \times 3 + 2 = 8$. C: $2 \times 2 + 5 = 9$.



3. Choose

C wins at score 9 among eligible activities.



4. Explain exclusion

B's score 11 cannot override its failed time requirement.

CHANGE ONE THING

Change interest weight from 2 to 3, keeping time limit 14.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Eligibility first

What happens and why: A and C fit time; B takes 15 and is excluded.

Step: 2. Scores

What happens and why: A: $2 \times 3 + 2 = 8$. C: $2 \times 2 + 5 = 9$.

Step: 3. Choose

What happens and why: C wins at score 9 among eligible activities.

Step: 4. Explain exclusion

What happens and why: B's score 11 cannot override its failed time requirement.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A score combines preferences according to explicitly chosen weights; it does not erase hard constraints. Computing a high score for an excluded item can be useful in an audit, but it must not allow that item back into the candidate set. Store both eligibility reasons and score components for explanations.

What this example does not establish

Toy interest scores are supplied inputs, not measured enjoyment or a universal definition of a good activity.

Read the labelled picture: Why does B not become the winner when its score is largest?

Change one thing: Change interest weight from 2 to 3, keeping time limit 14. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Weights are neutral facts rather than design choices. Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Weights represent preferences, not a discovered truth. Explain who chose them and test whether small

changes alter the recommendation. Avoid claiming that a larger score proves greater educational value.

For eligible A (fun 4, learning 1) and B (fun 2, learning 4), compare weights (2,1) with (1,2).

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Choosing the highest score and only then ignoring its duration breaks the requirement. Filter for eligibility first. If no item qualifies, report no eligible option instead of relaxing a requirement silently.

Practise with less help: Check a boundary

Use workbook W2 for the connected example “Weighted scores and constraints”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Weighted scores and constraints. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why does B not become the winner when its score is largest?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Create two reader profiles with different weights, calculate every term, total, rank, and explain why the order changes.

Create two reader profiles with different weights, calculate every term, total, rank, and explain why the order changes.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER

1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

- 1 • NAME: Define the concept without repeating the textbook.
 - 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
 - 3 • APPLY: Solve a different example independently.
 - 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
 - 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

Publish the features, weights, raw scores, and reason for the top result.

THIS WEEK’S CHECKPOINT Publish the features, weights, raw scores, and reason for the top result.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Publish the recommender’s constraints, score formula and tie rule; include component-level reasons in

each suggestion.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Create two user profiles and calculate book rankings.

CHALLENGE Create two user profiles and calculate book rankings.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 23 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Score=2 times genre match + length match. A matches both, B genre only, C length only.

The question

Score and rank.

Worked explanation

A=3, B=2, C=1. Weights encode a chosen preference, not universal reader truth.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Weighted preferences determine the score

Evidence or stage · Value or result

Evidence or stage: A

Value or result: 3

Evidence or stage: B

Value or result: 2

Evidence or stage: C

Value or result: 1

Score=2 times genre match + length match. A matches both, B genre only, C length only.

Score and rank.

A=3, B=2, C=1. Weights encode a chosen preference, not universal reader truth.

14. Try a new case independently

Independent case: Choose independently

Limit is 15 minutes. A has fun 3, learning 4, minutes 15; B has 5, 1, 16; C has 2, 5, 12. Calculate scores with $2 * \text{fun} + \text{learning}$ and choose.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 23 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A recommender gives 3 points for topic match and 1 for difficulty match. Book A matches topic only; B matches difficulty only. Compute scores and justify the ranking.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Publish the features, weights, raw scores, and reason for the top result.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Score= $3 \times \text{genreMatch} + 2 \times \text{short} + \text{illustrated}$. Compute $A=[1,0,1]$, $B=[0,1,1]$, $C=[1,1,0]$. A reader now requires illustrations. Which book wins after applying that constraint?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional book-activity chooser has a 14-minute limit.

By the end, I can · How I will show it

By the end, I can: Compute a weighted score

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply hard constraints before ranking

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Explain how weight choices change outcomes

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Compute a weighted score

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book-activity chooser has a 14-minute limit.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B's score 11 cannot override its failed time requirement.

1. Choose
2. Explain exclusion
3. Eligibility first
4. Scores

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Explain exclusion B's score 11 cannot override its failed time requirement. A score combines preferences according to explicitly chosen weights; it does not erase hard constraints. Computing a high score for an excluded item can be useful in an audit, but it must not allow that item back into the candidate set. Store both eligibility reasons and score components for explanations.

Practice 2: explain the deciding evidence

Changed-case practice: Change interest weight from 2 to 3, keeping time limit 14. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. A scores 11; C also scores 11; tie-by-ID chooses A.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must

say what follows from those facts.

After your attempt, compare with the supported explanation: A scores 11; C also scores 11; tie-by-ID chooses A. B remains too long; changing preferences changes only ranking of A and C.

Repair a missing building block

Review Nearest Neighbor Reasoning (ai-2/22). Goal: Compute absolute distances and rank training neighbours. Short practice: Why do $k=1$ and $k=3$ disagree for the original query?. Return to this week and explain what you changed.

Review Repair Bias in Data (ai-2/20). Goal: Identify selection and coverage bias. Short practice: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?. Return to this week and explain what you changed.

Review Noise and Outliers (ai-2/17). Goal: Calculate mean and median. Short practice: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.